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PAPERS (SHORT
AUTHORLIST)

* denotes joint first-author papers, † denotes student supervised.

30. †Neha Sharma, **Aditya Vijaykumar**, Prayush Kumar
Rapid inference of gravitational-wave signals in the time domain using a heterodyned likelihood
Submitted to PRD, [arXiv:2601.11239](#).
29. **Aditya Vijaykumar**, Amanda M. Farah, Maya Fishbach
The maximum mass ratio of hierarchical binary black hole mergers may cause the q - χ_{eff} correlation
Accepted in ApJL, [arXiv:2601.03457](#).
28. Amanda M. Farah, **Aditya Vijaykumar**, Maya Fishbach
The steep redshift evolution of the hierarchical binary black hole merger rate may cause the z - χ_{eff} correlation
Submitted to ApJL, [arXiv:2601.03456](#).
27. N.V. Krishnendu, Tamara Evstafyeva, ***Aditya Vijaykumar**, William E. East et al.
Implications of GW241011 for rotating exotic compact objects
Submitted to PRL, [arXiv:2511.17341](#).
26. Madison Van Wyngarden, Maya Fishbach, **Aditya Vijaykumar**, Alexandra G. Guerrero, Daniel E. Holz
How Low Can You Go: Constraining the Effects of Catalog Incompleteness on Dark Siren Cosmology
Submitted to ApJ, [arXiv:2511.04786](#).
25. Hui Tong et al. (including **Aditya Vijaykumar**)
Evidence of the pair instability gap in the distribution of black hole masses
Accepted in Nature, [arXiv:2509.04151](#).
24. Colm Talbot et al. (including **Aditya Vijaykumar**)
Inference with finite time series II: the window strikes back
Submitted to CQG, [arXiv:2508.11091](#).
23. Avinash Tiwari, Prolay Chanda, Shasvath J. Kapadia, Susmita Adhikari, **Aditya Vijaykumar**, Basudeb Dasgupta
Profiling Dark Matter Spikes with Gravitational Waves from Accelerated Binaries
Submitted to PRL, [arXiv:2508.03803](#).
22. Andris Doroszmai, Isobel M. Romero-Shaw, **Aditya Vijaykumar**, Silvia Toonen, et al.
Hierarchical Triples vs. Globular Clusters: Binary black hole merger eccentricity distributions compete and evolve with redshift
Submitted to MNRAS, [arXiv:2507.23212](#).
21. †Avinash Tiwari, **Aditya Vijaykumar**, Shasvath J. Kapadia, Shrobona Ghosh, Alex B. Nielsen
A pipeline to search for signatures of line-of-sight acceleration in gravitational wave signals produced by compact binary coalescences
Submitted to PRD, [arXiv:2506.22272](#).
20. Kanchan Soni, **Aditya Vijaykumar**, Sanjit Mitra
Assessing the potential of LIGO-India in resolving the Hubble Tension
Submitted to CQG, [arXiv:2409.11361](#).
19. †Avinash Tiwari, **Aditya Vijaykumar**, Shasvath J. Kapadia, Sourav Chatterjee, Giacomo Fragione
Profiling stellar environments of gravitational wave sources
[Phys. Rev. D 112, 084034](#), [arXiv:2407.15117](#).
18. Alexandra G. Hanselman, **Aditya Vijaykumar**, Maya Fishbach, Daniel E. Holz
Gravitational-wave dark siren cosmology systematics from galaxy weighting
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17. Sreejith Nair, ***Aditya Vijaykumar**, Sudipta Sarkar
Bounds on the charge of the graviton using gravitational wave observations
JCAP **11** (2024) 004, [arXiv:2405.05038](#).
16. ***Aditya Vijaykumar**, Alexandra G. Hanselman, Michael Zevin
Consistent eccentricities for gravitational wave astronomy: Resolving discrepancies between astrophysical simulations and waveform models
ApJ **969** 132, [arXiv:2402.07892](#).
15. Mukesh Kumar Singh, Shasvath J. Kapadia **Aditya Vijaykumar**, Parameswaran Ajith
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ApJ **971** 23, [arXiv:2312.07376](#).
14. [†]Kruthi Krishna, ***Aditya Vijaykumar**, Apratim Ganguly, *et al*
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13. **Aditya Vijaykumar**, Maya Fishbach, Susmita Adhikari, Daniel E. Holz
Inferring host galaxy properties of LIGO-Virgo-KAGRA's black holes
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12. Divyajyoti, N.V. Krishnendu, Muhammed Saleem, Marta Colleoni, **Aditya Vijaykumar**, K.G. Arun, Chandra Kant Mishra
Effect of double spin-precession and higher harmonics on spin-induced quadrupole moment measurements
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11. [†]Avinash Tiwari, ***Aditya Vijaykumar**, Shasvath J. Kapadia, Giacomo Fragione, Sourav Chatterjee
Accelerated binary black holes in globular clusters: forecasts and detectability in the era of space-based gravitational-wave detectors
MNRAS, **527**, 8586, [arXiv:2307.00930](#).
10. **Aditya Vijaykumar**, [†]Avinash Tiwari, Shasvath J. Kapadia, K.G. Arun, Parameswaran Ajith
Waltzing binaries: Probing line-of-sight acceleration of merging compact objects with gravitational waves
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In press: Astrobites
9. [†]Adhrit Ravichandran, **Aditya Vijaykumar**, Shasvath J. Kapadia, Prayush Kumar
Rapid Identification and Classification of Eccentric Gravitational Wave Inspirals with Machine Learning
Submitted to PRD, [arXiv:2302.00666](#).
8. Srashti Goyal, **Aditya Vijaykumar**, Jose Maria Ezquiaga, Miguel Zumalacarregui
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7. Bikram Keshari Pradhan, **Aditya Vijaykumar**, Debarati Chatterjee
Impact of updated Multipole Love and f-Love Universal Relations in context of Binary Neutron Stars
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6. **Aditya Vijaykumar**, Shasvath J. Kapadia, Parameswaran Ajith
Can a binary neutron star merger in the vicinity of a supermassive black hole enable a detection of a post-merger gravitational wave signal?
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5. **Aditya Vijaykumar**, Ajit Kumar Mehta, Apratim Ganguly
Detection and parameter estimation challenges of Type-II lensed binary black hole signals
Phys. Rev. D **108**, 043036, [arXiv:2202.06334](#).
4. Sumit Kumar, **Aditya Vijaykumar**, Alexander H. Nitz
Detecting Baryon Acoustic Oscillations with third generation gravitational wave observatories
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3. M. Saleem, Javed Rana, V. Gayathri, ***Aditya Vijaykumar** et al.
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2. **Aditya Vijaykumar**, M. V. S. Saketh, Sumit Kumar, Parameswaran Ajith, Tirthankar Roy Choudhury
Probing the large scale structure using gravitational wave observations of binary black holes,
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1. **Aditya Vijaykumar**, Shasvath J. Kapadia, Parameswaran Ajith
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PAPERS (LONG
AUTHORLIST,
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SUBSTANTIAL
CONTRIBUTION)

9. Abac et al. (LIGO Scientific, Virgo, and KAGRA Collaborations)
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7. Abac et al. (LIGO Scientific, Virgo, and KAGRA Collaborations) [**Paper Writing Team Lead**]
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[arXiv:2508.18083](#).
6. Abac et al. (LIGO Scientific, Virgo, and KAGRA Collaborations)
GWTC-4.0: Updating the Gravitational-Wave Transient Catalog with Observations from the First Part of the Fourth LIGO-Virgo-KAGRA Observing Run,
[arXiv:2508.18082](#).
5. Abac et al. (LIGO Scientific, Virgo, and KAGRA Collaborations)
GW231123: a Binary Black Hole Merger with Total Mass $190\text{--}265 M_{\odot}$,
[arXiv:2507.08219](#).
4. Abbott et al. (LIGO Scientific and Virgo Collaborations)
Tests of General Relativity with GWTC-3,
Accepted to *Physical Review D*, [arXiv:2112.06861](#).
3. Abbott et al. (LIGO Scientific and Virgo Collaborations)
Tests of General Relativity with Binary Black Holes from the second LIGO-Virgo Gravitational-Wave Transient Catalog,
Phys. Rev. D 103 (2021) 12, 122002, [arXiv:2010.14529](#).
2. Abbott et al. (LIGO Scientific and Virgo Collaborations)
GWTC-2: Compact Binary Coalescences Observed by LIGO and Virgo During the First Half of the Third Observing Run,
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1. P. Virtanen et al. (including **Aditya Vijaykumar** as SciPy 1.0 Contributor)
SciPy 1.0—Fundamental Algorithms for Scientific Computing in Python,
Nat Methods 17, 261–272 (2020), [arXiv:1907.10121](#).